Gateway PQC Re-Encryption Pipeline Report

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Source: `reencrypt_metrics.csv + firewall_misses.jsonl`

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1. Executive Summary

The gateway re-encrypted **605 benign messages** using `ML-KEM-512` + `ML-DSA-44` (AES-256-GCM). Mean broker transit was **13.202 ms** and mean PQC re-encryption overhead was **1.810 ms**.

Metric	Value
Messages re-encrypted	605
KEM algorithm	`ML-KEM-512`
Signature algorithm	`ML-DSA-44`
PQC mode	Real liboqs
Observation window	14:37:04 → 17:27:18

2. Timing Breakdown

> All values in milliseconds (ms). **Broker transit** = `t_recv_ns − t_publish_ns` (includes broker + network latency). **Total gateway** = full handler wall time from `on_message` entry.

Stage	N	Mean	Std	P95	Min	Max
Broker transit (client→gateway)	597	13.202 ms	71.045 ms	19.337 ms	0.173 ms	1543.454 ms
KEM encapsulate	605	0.441 ms	0.794 ms	0.916 ms	0.038 ms	10.912 ms
AES-256-GCM encrypt	605	0.119 ms	0.645 ms	0.172 ms	0.018 ms	15.712 ms
Signature sign	605	1.250 ms	1.847 ms	3.384 ms	0.049 ms	22.860 ms
Total re-encrypt (KEM+AES+Sign)	605	1.810 ms	2.248 ms	5.095 ms	0.112 ms	23.786 ms
Total gateway (on_message→done)	605	3.819 ms	11.579 ms	8.519 ms	0.318 ms	277.583 ms

3. Per-Device Timing

Device ID	N	Transit Mean	Transit Std	Reencrypt Mean	Reencrypt Std	Total GW Mean
benign-burst-dc2419	66	1.358 ms	0.574 ms	0.164 ms	0.044 ms	0.394 ms
benign-burst-f12ca2	194	14.019 ms	37.095 ms	2.613 ms	2.933 ms	4.787 ms
benign-mixed-fdf5f0	153	20.438 ms	126.499 ms	1.838 ms	1.445 ms	3.536 ms
benign-steady-59be62	30	1.444 ms	0.684 ms	0.176 ms	0.110 ms	0.576 ms
benign-steady-884c02	162	11.728 ms	41.394 ms	1.794 ms	2.054 ms	4.924 ms

4. Firewall Misses — Attacks That Reached the Gateway

Total missed attacks: 30

These messages had `is_attack=True` in the payload, meaning the attack client injected the label. The LightGBM firewall failed to classify or block the corresponding network flow before it reached the gateway. Use these records to identify model weaknesses.

By Attack Label

Attack Label	Count	Share
mqtt-flood	30	100.0%

By crypto_type Used (Spoofing Pattern)

crypto_type	Count
aes_gcm	30

Last 10 Miss Events

Time	Attack Label	Device	Seq	crypto_type	Payload Preview
17:22:19	mqtt-flood	flood-attacker	7112288	aes_gcm	{"device_id": "flood-attacker", "crypto_type": "aes_gcm", "s
17:22:20	mqtt-flood	flood-attacker	7112289	aes_gcm	{"device_id": "flood-attacker", "crypto_type": "aes_gcm", "s
17:22:21	mqtt-flood	flood-attacker	7112290	aes_gcm	{"device_id": "flood-attacker", "crypto_type": "aes_gcm", "s
17:22:22	mqtt-flood	flood-attacker	7112291	aes_gcm	{"device_id": "flood-attacker", "crypto_type": "aes_gcm", "s
17:22:23	mqtt-flood	flood-attacker	7112292	aes_gcm	{"device_id": "flood-attacker", "crypto_type": "aes_gcm", "s
17:22:24	mqtt-flood	flood-attacker	7112293	aes_gcm	{"device_id": "flood-attacker", "crypto_type": "aes_gcm", "s
17:22:25	mqtt-flood	flood-attacker	7112294	aes_gcm	{"device_id": "flood-attacker", "crypto_type": "aes_gcm", "s
17:22:26	mqtt-flood	flood-attacker	7112295	aes_gcm	{"device_id": "flood-attacker", "crypto_type": "aes_gcm", "s
17:22:27	mqtt-flood	flood-attacker	7112296	aes_gcm	{"device_id": "flood-attacker", "crypto_type": "aes_gcm", "s
17:22:28	mqtt-flood	flood-attacker	7112297	aes_gcm	{"device_id": "flood-attacker", "crypto_type": "aes_gcm", "s

5. Last 10 Re-Encrypted Messages

Device	Seq	Broker Transit	Re-Encrypt	Total GW
benign-burst-f12ca2	191	9.173 ms	2.246 ms	4.500 ms
benign-mixed-fdf5f0	151	8.381 ms	10.350 ms	11.662 ms
benign-burst-f12ca2	192	15.763 ms	2.356 ms	3.410 ms
benign-burst-f12ca2	193	14.828 ms	19.162 ms	27.323 ms
benign-burst-f12ca2	194	0.688 ms	1.337 ms	9.516 ms
benign-steady-884c02	160	5.707 ms	1.425 ms	2.650 ms
benign-mixed-fdf5f0	152	6.461 ms	0.880 ms	2.257 ms
benign-steady-884c02	161	5.891 ms	2.257 ms	3.630 ms

Device	Seq	Broker Transit	Re-Encrypt	Total GW
benign-mixed-fdf5f0	153	6.511 ms	0.985 ms	2.447 ms
benign-steady-884c02	162	5.999 ms	1.057 ms	2.282 ms

Report generated by `scripts/parse_gateway_reencrypt.py` — MQTT IDS/IPS + PQC Benchmarking Framework.